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Clinical Investigation Plan:

KISS Study (Keep bIfurcation Single stenting Simple)

Protocol no and/or	**”MED-04”**
“acronym”	“KISS”
Clinical Study Reference number	NCT04285372 (clinicaltrials.gov)
Version	3.0
Date	30 June 2021
Clinical Investigation Plan reference	**“MED-04-Version 3.0 (2021-06-30)”**
Sponsor	**CERIC** (CEnter for European Research Initiatives in Cardiovascular medicine) **3 rue des Granges – 1204 Geneva – SWITZERLAND**

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CLINICAL INVESTIGATION PLAN HISTORY

Version	Date	Main modifications
1.0	25/02/2020	Initial version
2.0	01/07/2020	Additional information on secondary analyses Additional information on the randomization process Term “observational study” modified in “current practice”
3.0	30/06/2021	Change of the sponsor’s logo Update of the name of the sponsor’s general manager Update on study timelines Additional information on the pre-procedure and post-procedure troponin measurements Update on statistical analysis Update on participating investigators and sites

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PROTOCOL SIGNATURE PAGE

PROTOCOL NUMBER: ”MED-04”

KISS Study (Keep bIfurcation Single stenting Simple)

I have read the Clinical Investigation Plan mentioned above and agree to adhere to its requirements.

I have received a copy of the most current version of the Medtronic Resolute Onyx Stent Instruction For Use.

I will provide copies of the Clinical Investigation Plan and access to all information furnished by CERIC (“Sponsor”) to the study personnel under my supervision and involved in carrying out the study. I will discuss this material with them to ensure that they are fully informed about the investigational device and the conduct of the study.

I have read the Confidentiality Statement of this Clinical Investigation Plan. The contents of this Clinical Investigation Plan may not be used in any other clinical study and may not be disclosed to any other person or entity without the prior written permission of the Sponsor. The foregoing shall not apply to disclosure required by law or regulation, for example submission to an Ethics Committee; however, I will give prompt notice to the Sponsor of any such disclosure.

Site Number: _____

Printed Name of Investigator: _____

Date: _____ Signature: _____

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Protocol Synopsis

Title	**KISS Study (Keep bIfurcation Single stenting Simple)**
Protocol Number	”MED-04”
Device	Medtronic Resolute Onyx stent
Study Design	Randomized, prospective, post-market, multicenter study
Study Objective(s)	To evaluate the non-inferiority of no side branch (SB) intervention versus side branch ballooning, in the setting of single stenting with systematic proximal optimization technique (POT), in the treatment of bifurcation lesion
Primary Endpoint(s)	Rate of peri-procedural Myocardial Infarction (MI) **at discharge or 48h post-procedure whichever comes first** as judged by clinical evaluation in association with day-1 troponin evaluation using the ARC-2 definition
Secondary Endpoints	**Predefined Secondary Procedural Endpoints:** • Technical success: deployment of main branch (MB) stent with <20% residual stenosis and a patent side branch at end of procedure upon corelab analysis • Total procedure time • Acute gain in main and side branches as measured with a dedicated quantitative coronary analysis (QCA) software • Fluoroscopy time • Total radiation dose (air kerma) • Total contrast dose • Need to cross over *(any intervention on SB in the test arm or failure to intervene on the SB in the control arm)* **Clinical endpoints** at **1 month** and **12 months:** Target Lesion Failure (TLF): Composite of: - Cardiac Death - Target vessel Myocardial Infarction - Clinically driven Target lesion revascularization (ci-TLR) • Main Branch & Side Branch TLR • Definite or Probable stent thrombosis (ARC-2 criteria)[1] • Angina status according to the Canadian Cardiovascular Society (CCS) classification (Grade 1 to 4)[2] *Within the SB intervention arm, additional analyses will be performed to compare SB ballooning vs Kissing Balloon Technique (KBT) subgroups* *[1]Garcia-Garcia HM, et al. Standardized End Point Definitions for Coronary Intervention Trials: The Academic Research Consortium-2 Consensus Document. Circulation. 2018 Jun 12;137(24):2635-2650.* *[2] Campeau L. Letter: Grading of angina pectoris. Circulation. 1976 Sep;54(3):522-3.*
Participating Sites	**Up to 20 sites over up to 6 European countries**
Study Population	**596 consecutive patients are planned to be enrolled**

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Inclusion Criteria	**Subject will be included if all of the following criteria are met:** 1. Coronary artery disease requiring revascularization that is amenable to percutaneous coronary intervention (PCI) 2. Any **de novo** Medina type bifurcation lesion with >70% lesion in the main vessel by visual assessment except Medina 0,0,1 if the planned strategy does not include a main branch stenting 3. Side branch diameter compatible with a PCI with stent implantation (≥ 2.25mm) 4. Side branch requiring a protection wire according to operator evaluation 5. Anatomy compatible with a proximal optimization technique 6. Post main branch treatment (stenting then POT): Absence of side branch occlusion Or low flow associated with ongoing ischemia (either chest pain or ST changes) 7. Male or female patient >18 years old
Exclusion Criteria	**Subject will not be included if any one of the following conditions exists:** 1. Acute myocardial infarction with ongoing ST-elevation 2. Cardiogenic shock 3. Requirement for ongoing hemodialysis 4. Life-expectancy limited to <12 months due to co-morbid condition 5. Lesion involves the left main coronary artery 6. Lesion requiring a double stenting 7. Plan to treat >1 other coronary vessel at the time of the index procedure 8. Two lesions treated in the same session: unsuccessful PCI on the single non bifurcation lesion in a non-target vessel before the treatment of the target lesion. 9. Chronic total occlusion of any target vessel 10. Left ventricular ejection fraction <20% 11. Side branch TIMI Flow <3 12. Pre-dilatation of the Side Branch during the procedure prior to randomization 13. Known allergy to Aspirin 14. Known allergy to Clopidorel and Prasugrel and Ticagrelor 15. Known allergy to stent drug elutant 16. Known allergy to any other component of Onyx Resolute stent 17. Ongoing participation in another investigational device or drug study 18. Inability to provide informed consent 19. Patients under judicial protection, tutorship or curatorship 20. Pregnant and breast-feeding women or intention to become pregnant prior to completion of all follow-up procedures
Risks and Benefits	**Expected benefits of no SB intervention:** • Facilitate bifurcation PCI procedure • Avoid ostium dissection and the need for a second stent • Reduce the risk of thrombosis and restenosis **Risks of no SB intervention:** • Side branch occlusion after MB stenting

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Statistical hypothesis/sample size/number of subjects	Considering alpha=0.05 et Beta=0.8 with a rate of 15% MI as defined by ARC-2 criteria in control group and a non-inferiority limit of 7.5%, 596 patients have to be randomized. Superiority will be tested if non-inferiority is met.
Duration of participation for each Subject	**Total duration for each subject is 12 months.** All enrolled subjects will be evaluated at 1 months and 1 year. All protocol required follow-up assessments must be conducted with direct contact with the subjects. *Protocol-required follow-up assessments may be performed via telephone interview or office visit as per local practice and subjects' mobility.*
Estimated duration of Inclusions	**Up to 24 months**
Total estimated duration of Study	**Up to 47 months** Duration of initiation: 6 Months Duration of inclusions: up to 24 Months Duration of follow-up: 12 Months Duration of close-out: 5 Months
Clinical Study Organization	**Sponsor**: CERIC (CEnter for European Research Initiatives in Cardiovascular medicine) 3 rue des Granges – 1204 Geneva – SWITZERLAND
	Monitor: CERC (Cardiovascular European Research Center) 7 rue du théâtre – 91300 Massy - FRANCE +33(0) 1 76 73 92 10 +33(0) 1 60 11 17 91
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Schedule of Assessments / Flow Chart:

Parameter/Examination	Screening	Day 1	1 months ± 7 days	12 months ± 1 month
Inclusion/exclusion	X			
Medical history	X		X	X
Angina status	X		X	X
Informed consent	X			
12-lead Electrocardiogram (ECG)	X	X		
PCI procedure	X			
Troponin level	X (Highly recommended: < 24h pre-procedure)	X (Between 6h and 48h post-procedure)		
Medication	X	X	X	X
Angiogram	X	X		
Adverse event monitoring		X	X	X

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List of Abbreviations

ACCF	American College of Cardiology Foundation
ACT	Activated Clotting Times
AHA	American Heart Association Task Force on Practice Guidelines
AE	Adverse Event
AT	As-treated
ARC	Academic Research Consortium
BARC	Bleeding Academic Research Consortium
BMS	Bare Metal Stent
CA	Competent Authority
CABG	Coronary Artery Bypass Graft
CAD	Coronary Artery Disease
CCS	Canadian Cardiovascular Society
CE	Conformité Européenne
CEC	Clinical Events Committee
CERC	Cardiovascular European Research Center
CK	Creatine Kinase
CK-MB	Creatine Kinase-Myocardial Band
CIP	Clinical Investigation Plan
ci-TLR	Clinically driven Target Lesion Revascularization
CRA	Clinical Research Associate
CRF	Case Report Form

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CRO	Clinical Research Organization
DAPT	Dual Anti-Platelet Therapy
DCS	Drug Coated Stent
DMSO	DiMethyl SulfOxide
EACTS	European Association for Cardio-Thoracic Surgery
EC	Ethics Committee
ECG	Electrocardiogram
e-CRF	electronic Case Report Form
EDC	Electronic Data Capture
ESC	European Society of Cardiology
FDA	U.S. Food and Drug Administration
GDPR	European General Data Protection Regulation
hs-cTn	High-sensitivity cardiac troponin
ICF	Informed Consent Form
ICJME	International Committee of Medical Journal Editors
ID	Identifier
IFU	Instructions For Use
IRB	Institutional Review Board
ITT	Intent-to-treat
IVUS	IntraVascular UltraSound
KBT	Kissing Balloon Technique
LBBB	Left Bundle Branch Block
MB	Main Branch

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MI	Myocardial Infarction
OAC	Oral Anticoagulant
OCT	Optical Coherence Tomography
OTW	Over-the-Wire
PCI	Percutaneous Coronary Intervention
PI	Principal investigator
PIC	Patient Informed Consent
POT	Proximal Optimization Technique
PP	Per-Protocol
PVP	PolyVinyl Pyrrolidone
QCA	Quantitative Coronary Analysis
RBC	Red Blood Cells
RX	Rapid Exchange
SAP	Statistical Analysis Plan
SAPT	Single Anti-Platelet Therapy
SB	Side Branch
SCAI	Society for Cardiovascular Angiography and Interventions
SOP	Standard Operating Procedures
STEMI	ST segment Elevation Myocardial Infarction
TIMI	Thrombolysis In Myocardial Infarction
TLF	Target Lesion Failure
TLR	Target Lesion Revascularization
TVR	Target Vessel Revascularization

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1. STUDY RATIONALE/STUDY JUSTIFICATION

Treatment of coronary bifurcation lesions is a complex intervention although relying on angioplasty, a technique routinely used by interventional cardiologists. The particularity resides in the necessity of a sequential approach on the main branch and the side branch using either 1 stent for the MB or 2 stents for both the MB and the SB.

Still, with similar therapies, while bifurcation lesions represent 15 to 20% of PCIs, those procedures are associated to lower success rates and increased complications occurrences.(1–4) Studies conducted during the last decade evaluating 1 vs 2 stents strategies in the treatment of bifurcation lesions observed better 5-years outcomes reducing patient mortality rate when using a single pharmacologically active stent (Drug coated stent – DCS) approach.(5–11).

However, while recent studies have proven the benefit single stenting in the treatment of bifurcation lesion(7), the debate is still there around the added value of SB interventions in the setting of a provisional stenting approach when conducted according to European Bifurcation Club recommendations.(12) Thus, the role of side branch intervention for improving the safety of PCI, when MB is treated according to the Murray law with a POT technique, remains unclear.(13) Studies evaluating the potential benefit of Kissing Balloon Technique (KBT) have been conducted before the systematic use of POT. Registries and guidelines promote the use of KBT when a two-stent technique is applied.(14,15) More recently, bench testing and imaging study has suggested that side branch ballooning followed by a second POT could be an alternative to KBT.(12,16–19)

However, studies have shown that SB angioplasty is associated with an increased risk of dissection leading to a higher rate of double stenting.(20–24)

In the treatment of bifurcation lesion, the KISS study will evaluate the non-inferiority of no side branch (SB) intervention versus side branch ballooning, in the setting of single stenting with systematic proximal optimization technique (POT), in the treatment of bifurcation lesion.

2. INVESTIGATIONAL DEVICE(S) – IDENTIFICATION AND DESCRIPTION

The Medtronic Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System (Resolute Onyx™ system) is a device/drug combination product comprised of the following device components: the Resolute Onyx™ coronary stent and delivery system and a drug component (a formulation of zotarolimus in a polymer coating).

Medtronic Resolute Onyx stent obtained its CE (Conformité Européenne) mark in 2014 and a U.S Food and Drug Administration (FDA) Approval in 2017 (25). Complementary information to the following sections can be found in Medtronic Resolute Onyx Stent Instructions for Use (IFU).(26)

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2.1 DEVICE COMPONENT DESCRIPTION

The Medtronic Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System (Resolute Onyx™ system) consists of a balloon-expandable, intracoronary, drug-eluting stent (DES) premounted on a Rapid Exchange (RX) or an Over-the-Wire (OTW) stent delivery system.

The Resolute Onyx™ stent is manufactured from a composite material of cobalt alloy and platinum-iridium alloy and is formed from a single wire bent into a continuous sinusoid pattern and then laser fused back onto itself. The stents are available in multiple lengths and diameters. The delivery system has two radiopaque markers to aid in the placement of the stent during fluoroscopy and is compatible with 0.014-inch (0.36-mm) guidewires and 1.42-mm (5-Fr/0.056-in) minimum inner diameter guide catheters. The Resolute Onyx™ RX delivery system and the Resolute Onyx™ OTW delivery system have an effective length of 140 cm. (26)

2.2 DRUG COMPONENT DESCRIPTION

The drug coating of Resolute Onyx™ System consists of the drug Zotarolimus (the active ingredient) and BioLinx® polymer system (the inactive ingredient). (26)

Zotarolimus
The active pharmaceutical ingredient utilized in the Resolute Onyx™ System is zotarolimus. It is a tetrazole-containing macrocyclic immunosuppressant. Zotarolimus has extremely low water solubility and is a lipophilic compound that is freely soluble in Propylene glycol, Acetone, Toluene, Acetonitrile, Ethanol, Benzyl alcohol and Dimethyl sulfoxide (DMSO). The molecular formula of zotarolimus is C52H79N5O12 and its molecular weight is 966.2. Zotarolimus does not have any ionizable group(s) in the physiological pH range; therefore, its solubility is expected to be unaltered in this range.

Polymer System Description
The Resolute Onyx™ stent is comprised of a bare metal stent (BMS) with a Parylene C primer coat and a coating that consists of a blend of the drug zotarolimus and the BioLinx® polymer system. BioLinx® is a blend of the Medtronic proprietary components C10 and C19, and PVP (polyvinyl pyrrolidone).

2.3 INDICATIONS

The Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System is indicated for improving coronary luminal diameters in patients, including those with diabetes mellitus, with symptomatic ischemic heart disease due to de novo lesions of length ≤ 35 mm in native coronary arteries with reference vessel diameters of 2.25 mm to 5.0 mm.(26)

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2.4 CONTRAINDICATIONS

The Resolute Onyx™ System is contraindicated for use in: (26)

- Patients with known hypersensitivity or allergies to aspirin, heparin, bivalirudin, clopidogrel, prasugrel, ticagrelor, ticlopidine, drugs such as zotarolimus, tacrolimus, sirolimus, everolimus, or similar drugs or any other analogue or derivative.
- Patients with a known hypersensitivity to the cobalt-based alloy (cobalt, nickel, chromium, and molybdenum) or platinum-iridium alloy.
- Patients with a known hypersensitivity to the BioLinx® polymer or its individual components

Coronary artery stenting is contraindicated for use in: (26)

- Patients in whom anti-platelet and/or anticoagulation therapy is contraindicated.
- Patients who are judged to have a lesion that prevents complete inflation of an angioplasty balloon or proper placement of the stent or stent delivery system.

3. STUDY OBJECTIVE AND ENDPOINTS

The primary objective of KISS study is to evaluate the non-inferiority of no side branch (SB) intervention: **test arm**, versus side branch ballooning: **control arm**, in the setting of single stenting with systematic proximal optimization technique (POT), in the treatment of bifurcation lesion.

3.1 PRIMARY ENDPOINT(S)

Rate of peri-procedural MI **at discharge or 48h post-procedure** whichever comes first as defined by the ARC consortium (Academic Research Consortium, so called “ARC-2” definitions)(27) and based on post-PCI troponin rise as follows:

Periprocedural MI PCI (within 48 Hours)**:**

Absolute rise in cardiac troponin (hs-cTn: high-sensitivity cTn) from baseline ≥35 times upper reference limit

+/and one (or more) of the following criteria:

- New significant* Q waves or equivalent
- Flow-limiting angiographic complications
- New “substantial” loss of myocardium on imaging

Significant periprocedural myocardial injury:

Absolute rise in cardiac troponin (hs-cTn) from baseline ≥70 times upper reference limit

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*These definitions are applicable in patients with normal (or elevated and stable or falling) baseline biomarkers. *Q-wave criteria requires the development of new Q waves ≥40 ms in duration and ≥1 mm deep in voltage in ≥2 contiguous leads.*

3.2 Secondary Endpoints

3.2.1 Procedural Endpoints:

- Technical success: deployment of main branch stent with <20% residual stenosis and a patent side branch at end of procedure upon Corelab analysis (additional information available in the Corelab Plan)
- Total procedure time
- Acute gain in main and side branches as measured with a dedicated QCA software
- Fluoroscopy time
- Total radiation dose (air kerma)
- Total contrast dose
- Need to cross over *(any intervention on SB in the test arm or failure to intervene on the SB in the control arm)*

3.2.2 Clinical endpoints:

Rates of the following endpoints will be evaluated at 1-month and 12-months:

- Target Lesion Failure (TLF) defined as the composite of:
  - Cardiac Death,
  - Target vessel Myocardial Infarction,
  - Clinically driven Target lesion revascularization (ci-TLR).

  Target lesion is the segment where coronary intervention was performed, and the length of the target lesion is inclusive of the arterial segment treated with the stent and the 5mm proximal and 5mm distal to the treated section.
- Main Branch (MB) & Side Branch (SB) TLR
- Definite or Probable stent thrombosis (ARC-2 criteria)
- Angina status according to the CCS classification (Grade 1 to 4) (28)

4. Risk-Benefit Analysis

4.1 Risks

The Medtronic Onyx stent used in this trial received CE mark approval based on risk assessment after bench testing, pre-clinical testing and first clinical applications supporting the safety and efficacy of the device. For more detailed information on intended use of the device, indications and contraindications, as well as a complete list of warnings, precautions

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and potential adverse effects, please refer to the IFU of the Medtronic Resolute Onyx stent, which are provided with the product.(26)

As this study is performed as current practice, additional risks are those associated to the routine implantation of a coronary stent in a native coronary artery in other words risks related to conventional percutaneous transluminal coronary angioplasty. Moreover, risks factors may differ from one patient to another and risks evaluation should be mitigated by assessing each intervention complexity and patient morbidity criteria.(26)

Specifically, the expected risks in both KISS randomisation arms are those identified from the literature,(1,2,12–14,18,19,23,24,26,29) namely:

- no intervention on the SB group: risk of SB occlusion,
- SB ballooning group: risk of SB dissection that would require a second stent.

4.2 BENEFITS

The main benefit of this study is to explore the best practice for future patients in treating bifurcations lesions with a main branch single stent with or without side branch intervention strategy.

Hence this trial will contribute to expand the knowledge on the treatment of coronary bifurcation lesion. An additional benefit of this trial includes patient exposure to Good Clinical Practice standards.

Patients will be treated with the Medtronic Onyx stent which gained CE mark and is commercially available following to favorable procedural and clinical outcomes since 2014. Therefore, patients in this trial will undergo thorough pre-procedural and follow-up assessments with the same level of care as a standard routine bifurcation PCI procedure.

Careful monitoring of the patients after the procedure to rapidly detect and manage any complication is standard-of-care for all patients undergoing a PCI procedure independent of KISS study participation. Based on the above considerations and the current status of scientific evidence, we regard the risk-benefit relation to be in favor of conduction of this trial.

5. STUDY DESIGN

5.1 MAIN CONSIDERATIONS

KISS study is an investigator-initiated, multi-centre, prospective, randomized (1:1), parallel, two-arm, non-inferiority trial aiming to compare two bifurcation PCI procedures for SB protection, performed in up to 6 countries where Medtronic Resolute Onyx stent is CE approved.

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Five hundred ninety-six (596) consecutive patients are planned to be enrolled in up to 20 participating investigational sites.

5.2 SUBJECTS

KISS study is intended to include patients with coronary heart disease particularly patients with **de novo** coronary bifurcation lesion with >70% lesion in the main vessel requiring revascularization.

There should be no more than 1 additional coronary lesion (excluding a bifurcation lesion) not on the target vessel requiring treatment, such that the main focus of the study will rest on the bifurcation disease. The lesion which is to be treated with PCI should be treated preferably before, the index lesion. In the case of stable patients this will allow randomisation to occur after treatment of non-index lesion.

If other lesions are to be treated, drug-eluting stents (Medtronic Resolute Onyx) are to be used to reduce risk of subsequent revascularisation unrelated to the index bifurcation disease.

5.2.1 Inclusion Criteria

Subject will be included if all of the following criteria are met:

1. Coronary artery disease requiring revascularization that is amenable to PCI
2. Any **de novo** Medina type bifurcation lesion with >70% lesion in the main vessel by visual assessment except Medina 0,0,1 if the planned strategy does not include a main branch stenting
3. Side branch diameter compatible with a PCI with stent implantation (≥ 2.25mm)
4. Side branch requiring a protection wire according to operator evaluation
5. Anatomy compatible with a proximal optimization technique
6. Post main branch treatment (stenting then POT): Absence of side branch occlusion Or low flow associated with ongoing ischemia (either chest pain or ST changes)
7. Male or female patient >18 years old

After the POT and in the absence of SB occlusion or SB low flow associated with ongoing ischemia (either chest pain or ST changes), patient will be randomized to one or other procedure arm.

5.2.2 Exclusion Criteria

Subject will not be included if any one of the following conditions exists:

1. Acute myocardial infarction with ongoing ST-elevation
2. Cardiogenic shock

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3. Requirement for ongoing hemodialysis
4. Life-expectancy limited to <12 months due to co-morbid condition
5. Lesion involves the left main coronary artery
6. Lesion requiring a double stenting
7. Plan to treat >1 other coronary vessel at the time of the index procedure
8. Two lesions to be treated in the same session: unsuccessful PCI on the single non-bifurcation lesion in a non-target vessel before the treatment of the target lesion
9. Chronic total occlusion of any target vessel
10. Left ventricular ejection fraction <20%
11. Side branch TIMI Flow <3
12. Pre-dilatation of the Side Branch during the procedure prior to randomization
13. Known allergy to Aspirin
14. Known allergy to Clopidorel and Prasugrel and Ticagrelor
15. Known allergy to stent drug elutant
16. Known allergy to any other component of Onyx Resolute stent
17. Ongoing participation in another investigational device or drug study
18. Inability to provide informed consent
19. Patients under judicial protection, tutorship or curatorship
20. Pregnant and breast-feeding women or intention to become pregnant prior to completion of all follow-up procedures

Furthermore, for positively screened patients receiving a bifurcation PCI, after the POT and in the presence of SB occlusion or SB low flow associated with ongoing ischemia (either chest pain or ST changes), patient will be excluded.

5.3 ENROLMENT PROCESS

Consecutive patients planned for bifurcation PCI will be screened for eligibility. If patients fulfil all inclusion and do not meet exclusion criteria, they will be informed about the study`s purpose and course and will be asked for participation and written informed consent.

All the details of the study will be explained in the Informed Consent Form (ICF) (Appendix I).

After all eligibility criteria for the study have been met, including that a treatable lesion has been identified, predilated and stented, the post-Main Branch stenting strategy will be determined by the randomization process centrally performed by blocks with site stratification.

To limit selection bias and any potential confounding factor, patients will be randomized 1:1 using KISS e-CRF during the PCI after main branch stenting to either strategies:

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- no intervention on the Side Branch,
- Side Branch ballooning or Kissing Balloon Technique - KBT (left at operator discretion) using a non-compliant 1:1 balloon

5.4 SUBJECT PARTICIPATION AND FOLLOW-UP PERIOD

Patients inclusion period will be up to 24 months. The total participation follow-up duration for each subject is expected to be of 12 months. Patient will be assessed at the following intervals: baseline, procedure, discharge, 1 month, and 12 months.

The overall study duration will be up to 47 months (study initiation: 6 months; recruitment up to 24 months; follow up:12 months; close-out: 5 months).

Each center should include at least one patient, but not more than 30% of the total number of patients.

5.5 PATIENT WITHDRAWAL

Subjects will be encouraged by investigators to remain in the trial until completion of the 12-months follow-up if reasonable according to investigator's decision. However, subjects are free to withdraw consent at any time, without having to justify, and without influence on their further medical treatment or relationship to their physicians. Once subjects are withdrawn, they will be followed as per institutional standard of care and no more data will be collected in the e-CRF.

Subjects 1) who have signed an informed consent form, but have been declared ineligible for the study, 2) who were declared eligible for the study but are discontinued for any reason prior to the commencement of any procedure; will be considered as screen failed and documented as such in the e-CRF and the patient flow-chart. Data until that time period will be collected and they will be immediately withdrawn from study. They will be considered as screen failed and documented as such in the e-CRF and the patient flow-chart. After study exit, they will then be treated per the discretion of Investigators; however, no study material may be used and no follow-up data will be collected in the e-CRF

For all the withdrawn subjects, the reason for the early withdrawal must be documented on the e-CRF study exit form. They will be recorded as informed consent withdrawal or study withdrawal and declared accordingly in the patient flow chart.

If a subject withdraws from the study, he will be censored at the time when the last follow-up interview or examination took place. Upon study withdrawal, subjects will be followed per normal standard of care.

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5.6 DISCONTINUATION CRITERIA

If the Sponsor, Regulatory/Competent Agency/Authority, and/or the Principal Investigator (PI) discover conditions during the study that indicate the investigation should be terminated for patient safety reasons, an appropriate schedule for termination will be setup.

Discontinuation may occur at subject's own request.

5.7 SUBJECT IDENTIFICATION AND CONFIDENTIALITY

Subjects will be identified on the e-CRF and any other document transmitted to the Sponsor by the PI or clinical site staff, by a unique identification number to maintain patients confidentiality throughout the clinical study. No personal data will be collected in the e-CRF and patients are identified by a unique patient identification code (site ID – patient ID).
Pseudo-anonymized data entered on the e-CRF or part of the imaging are confidential and will only be available to the Sponsor (including Sponsor delegates), members of data management teams, the statistician, Clinical Events Committee (CEC) members, members of the ECs and if requested to CAs.
All documents sent CERC or to the Core Lab should be pseudo-anonymized by the site.
The PI for each clinical investigation site will maintain as part of the investigational file a list identifying all subjects enrolled into the clinical investigation. Subjects who are enrolled (signed informed consent) and found ineligible shall be documented on this enrolment log.

6. STUDY HANDLING PROCEDURES

6.1 PRE-PROCEDURE

Medications:
Patients shall be treated according to local standard or care. Following provided information summarize actual therapeutic recommendations.

Prior to PCI, and according to literature including ACCF/AHA/SCAI(30) and ESC/EACTS(31) Guidelines and study stent manufacturer(26), all patients shall be treated with dual antiplatelet therapy (DAPT). According to these recommendations:

If the patient has not been established on Aspirin 75mg daily for at least 5 days prior to the PCI, a loading dose of 250 mg may be given on the morning of the PCI. Combined dual antiplatelet treatment with Clopidogrel, Prasugrel or Ticagrelor are acceptable and may be directed by the attending Physician. If either Clopidogrel 75mg daily or Prasugrel 10mg daily or Ticagrelor 90mg twice daily has not been taken for at least 5 days prior to the PCI, then a loading dose of the drug may be administered. Recommended loading doses are :

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Clopidogrel 600mg, Prasugrel 60mg and Ticagrelor 180mg at least 2 hours pre-procedure.(30,31)

Clinical assessment:
Prior to the revascularisation procedure, all patients will undergo an assessment of their medical history and their angina status (CCS classification)(28). Routine blood tests will be performed as per local practices including patient troponin levels, for which it is recommend being assessed not more than 24h prior to the procedure, as well as a 12-lead ECG. Baseline angiogram will be collected in the e-CRF.

6.2 PROCEDURE

PCI:
As this study is performed as current practice, PCI may be undertaken via the access site of choice of the operator. All following information are indicative for investigators.

Haemostatic technique and use of vascular closure devices will also be at the discretion of the operator. Target Activated Clotting Times (ACTs) during the procedure should be >250 seconds. A PCI on a single non-bifurcation lesion in a non-target vessel is possible in the same session if successfully performed before the treatment of the target lesion.

The patient will undergo a bifurcation PCI procedure, where the preferred procedural strategy, is as follows:

- Both limbs of the bifurcation should be wired after two baseline angiographic views with minimal overlap between branches and clear visualization of SB ostium. After an appropriate preparation of the MB lesion (left at the operator discretion), the stent (Onyx Resolute) should be chosen with 1:1 diameter ratio calculated on the distal MB reference diameter and a length allowing an implanted stent length in the proximal main branch compatible with a POT using 1:1 balloon/artery ratio.
  Please note that any SB predilatation during the procedure will result in patient exclusion.

- After the POT and in the absence of SB occlusion or SB low flow associated with ongoing ischemia (either chest pain or ST changes), patient will be randomized to one or other procedure arm (See Section 5.3). A wire should remain jailed in the non-stented vessel until randomization.

- A final POT is recommended in the KBT group, but this second (final) POT is mandatory if a single ballooning has been performed in the side branch (so-called “POT-Side-POT” technique).

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Adjunctive imaging (Intravascular ultrasound - IVUS or optical coherence tomography - OCT) is possible. IVUS runs should be recorded and made available to potential Core Laboratory analysis.

The two wires should be removed before repeating the two final angiographic views using the same projections than baseline angiographies.

6.3 POST-PROCEDURE

Patients follow-up will consist in 2 clinical visits at 1 month and 12-months. All protocol required follow-up assessments must be conducted with direct contact with the subjects as per local practice and subjects' mobility.

In case of occurrence of clinical endpoints or adverse events, inquiries at the treating family practitioner or specialist physicians will be conducted and corresponding medical records consulted. Subjects will be followed up until completion of the 12 months follow-up visit. Upon completion of the final protocol assessment, subject participation will be considered complete and subjects will be followed as per local standard of care.

A core laboratory will be used to perform quantitative coronary analysis (QCA) on endpoints requiring a re-PCI. This is done prior to sending the baseline and second angiogram to the CEC members for adjudication of the endpoint.

Post PCI, it is recommended that the patient continues on Aspirin and either Clopidogrel or Prasugrel or Ticagrelor in accordance with current ESC guidelines.(31)

6.3.1 Day 1 and hospitalization

During the hospitalization prior to hospital discharge, standard tests and/or assessments will be performed as per local practice and corresponding information recorded: Physical examination, monitoring of vital signs, routine blood tests, changes in medications…

Particular attention should be paid to primary endpoint related assessments particularly 12-lead ECG within 24 hours after the procedure and routine blood tests including **troponin levels** assessment at least 6 hours and not later than **48 hours** after the procedure **for periprocedural MI detection**.(27)

In case of multiple troponin values, sites are requested to provide the peak values (12-48 hours).

6.3.2 Discharge

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Anti-thrombotic/anticoagulation therapy should be continued after discharge as recommended by the ACCF/AHA/SCAI(30) and ESC/EACTS(31) guidelines. As a reminder, DAPT is generally recommended for 6 months followed by single anti-platelet therapy for life. The remaining medications (including oral anti-coagulants – OACs) is left upon discretion of the treating physician.

The investigator shall ensure that upon discharge, all primary and secondary endpoints related tests and assessments were performed and recorded on the e-CRF (See sections 3.1 and 3.2).

6.3.3 1-month and 12 months follow-up visits

For patient follow-up, clinical visits are encouraged for a patient physical examination to be performed and facilitate discussions. However, whenever these visits are performed by phone, all efforts should be made by the investigator to ensure collection of validated and documented follow-up data.

Following tests and/or assessments shall be performed as per standard of care during the 1-month and the 12-months follow-up visits:

- Target Lesion Failure (TLF), defined as the composite of:
  - Cardiac Death
  - Target vessel Myocardial Infarction
  - Clinically driven Target lesion revascularization (TLR)
- Angina status as defined by the CCS classification(28)
- Review of medical history,
- Physical examination (unless the follow-up visit is conducted by phone),
- Record of medication use,
- Occurrence of any endpoint related event,
- Occurrence of any protocol deviation.

6.4 MONITORING PLAN

The Sponsor is responsible for ensuring appropriate monitoring of the clinical investigation activities. This task has been delegated by the Sponsor to CERC who appointed monitors independent from the investigators and Sponsor. The monitors will be the primary contact for the PI and clinical investigation site personnel. Their responsibilities include collecting and tracking data forms and instituting quality control measures for data entry verification and study compliance.

Consequently, the KISS monitoring plan shall be written to provide the frequency of the monitoring visits, tasks that will be performed as well as compliance rules such as provisional corrective actions for any detected non-compliance by the sites. Further predefined quality

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tolerance limits shall be defined with related triggers for risk assessment to determine if actions are needed to improve compliance as soon as threshold are reached or exceeded (risk-based monitoring).

To ensure adequate study conduct, training will be done on an individual basis during a site visit before the first enrollment at each participating center. Additional training may be provided as needed. A training log will be used to document that training has been done. This will be filed in the site binder. When a new member joins the study site staff, the PI/delegate or CRO monitors will provide the appropriate training prior to their performing any study related activity.

Data will be collected by study Investigators and reported on standard data collection forms or Case Report Form (CRF) provided by the Sponsor. Data entry will be performed in a manner that ensures accuracy. The entered data will be audited for verification and validation purposes. The CRFs will be reviewed for errors, omissions, internal consistency, and to ensure that the investigator has signed and dated the appropriate sections. Data collection is detailed in section 9 and based on the study endpoints outlined above.

Periodic monitoring visits will be made at the study site throughout the clinical study and in accordance with the Monitoring Plan, to assure that all investigators conduct the study in compliance with the clinical investigation plan (CIP), the Investigators' agreements and all applicable regulations and guidelines. These visits will further assure that the facilities and technical equipment are still acceptable, all members of site staff well identified and trained, the EC/CA has been notified of approved protocol changes as required, complete records are being maintained, appropriate and timely reports have been made to the Sponsor and the EC/CA, device and device inventory are controlled. The clinical sites shall permit and assist the monitor to carry out all required on site verifications, they will be monitored to ensure that data are entered into the e-CRF in a timely manner and that the completed e-CRFs match the medical records per the Monitoring Plan (source data verification), and resolve any differences.

At the end of the study, there will be a close-out visit performed at each site to ensure that the PI's records are complete, all documents needed for the Sponsor's files are retrieved, remaining clinical investigation materials are disposed of or returned, previously identified issues have been resolved and all parties are notified including the CA (if applicable).

7. ADVERSE EVENT MANAGEMENT

As this study is performed as current practice, no procedure of management of serious adverse device effect is imposed to investigators, regarding collection in the e-CRF. Nonetheless, declaration of serious adverse device effect is mandatory as usual practice and must be done to the local structure in the centre dedicated to device vigilance.

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For the purpose of the study, all endpoint-related adverse events will need to be reported in the e-CRF.

7.1 Reference Safety Information

As a commercially available and CE marked Device, all the safety information related to the Medtronic Resolute Onyx stent are available in the Instruction for Use document (26).

7.2 Adverse Event Reporting

Endpoint related events will be reported and documented in the e-CRF. As much information as possible should be gathered with regard to the adverse event.

The occurrence of all endpoint-related events should be made available to the CEC to be adjudicated and reviewed in blinded form with regard to treatment allocation. Subjects will be identified in the reports according to their unique identification code assigned after enrolment and randomization. The Steering Committee, Sponsor, or the EC may request additional information from the responsible investigator.

For all these events, the causal relationship with the investigational device or procedure should be established by the Investigator as detailed as follow:

- Unrelated: the event is definitely not associated with the procedures
- Possible: the temporal sequence between the procedures and the event is such that the relationship is likely or patient's condition or concomitant therapy could have caused the adverse event (AE)
- Probable: the temporal sequence between the procedure and the event is relevant or the event abates upon procedures completion/removal (dechallenge) or the event cannot be reasonably explained by the patient's condition
- Definite / Certain: the temporal sequence is relevant and the event abates upon procedures completion/removal (dechallenge) or reappearance of the event on repeated procedure (rechallenge)

The information for the event should include the date of onset and resolution, the action taken, the corrective treatment and how the subject recovered with or without sequelae.

In case of death, whether the death is related to the investigational device and/or the study procedure must be well documented. The date on which subject died, what attempts were made to treat the event that led to death, the performance and functioning of the investigational device during the event should be noted.

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For all events that will be adjudicated by the Clinical Event Committee, source documents (medical records and reports…) will be provided to clinical research associates (CRAs).

Minor adverse events will not be recorded in the e-CRF.

7.3 Follow-Up of Adverse Events and Adverse Device Effects

Any declared AE must be followed until it has resolved, has a stable level of sequelae or in the Investigator's opinion is no longer clinically significant.

8. Statistical Considerations

8.1 Sample size justification

Assuming a rate of 15% Myocardial Infarction as defined by ARC-2 criteria in the control group, taking a 6% drop out rate into account and using an absolute non-inferiority limit of 7.5%, 596 subjects randomized equally between the 2 arms will have 80% power to show non-inferiority at a one-sided significance level of 0.05.

8.2 Analysis Sets

Three analysis sets will be defined. The primary analysis set is the intention-to-treat (ITT) set which includes all patients randomized. Patients in the ITT set will be analysed based on the procedure arm to which they were originally allocated to, irrespective of any crossing-over or treatment protocol violations.

The second analysis set is the per-protocol (PP) set, which includes all patients randomized who did not had any serious treatment protocol violations. The PP set will be determined on a blinded (to outcome) data review meeting before data base lock. Patients in the PP set will be analysed based on the procedure arm to which they were originally allocated to.

The third analysis set is the "as-treated" (AT) set, which includes all patients randomized and who underwent the procedure. Patients in the AT set are analysed in the treatment group of which procedure actually was received.

8.3 Primary analyses

The primary analyses will be performed on the ITT and PP set. The AT set will serve for sensitivity analyses.

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The primary endpoint will be compared between both groups by Farrington-Manning Score Test. The difference between two treatment groups will be reported with a one-sided 95% confidence interval. In case of missing data, multiple imputation methodology with 100 imputations will be performed. Specific details will be specified in the statistical analysis plan.

Non-inferiority should only be claimed if both the ITT and the PP analyses show a difference in the primary endpoint with an upper limit of a one-sided 95% confidence interval below the non-inferiority margin of 7.5%.

In case non-inferiority is met for the primary endpoint, a superiority analysis will be performed using a two-tailed significance level alpha of 0.05.

8.4 SECONDARY ANALYSES

Secondary analyses will be observationally performed. To this aim, differences between the two arms regarding the secondary endpoints will be evaluated for hypotheses generating purposes and future studies development. Difference between the two arms regarding secondary endpoints including Target Lesion Failure (TLF) and angina status (section 3.2 Secondary endpoints) will be analysed at 1-month and 12-months in intention-to-treat and per protocol basis.

Comparison between the two treatment groups will be performed on demographical, clinical, procedural and imaging characteristics. Continuous variables will be presented as mean ± standard deviation or median and interquartile range depending on their distribution. Categorical variables will be stated as number and proportion.

Further secondary analyses will encompass secondary clinical endpoints at 30 days and 12 months and pre-specified subgroup analyses like to sex, coronary artery disease status, etc. with appropriate interaction tests.

Full details of all analyses will be included in a statistical analysis plan (SAP) which will be finalized before data base lock.

8.5 HANDLING OF MISSING DATA AND DROP-OUTS

If the primary endpoint is missing, multiple imputation methodology with 100 imputations will be performed. Specific details about the imputation method will be specified in the statistical analysis plan.

For time to event endpoints, if a patient withdraws from the study or is lost to follow-up, the subject will be censored at the time when the last follow-up examination took place

For other endpoints, no imputation is foreseen.

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9. DATA COLLECTION AND DATA HANDLING

Subject data will be collected through a secure internet-based electronic data capturing (EDC) system provided to the centres prior to study start. This system will be used to record all subject information collected in the clinical investigation for secure data tracking and centralized data monitoring ("remote monitoring"). Automatic queries will be built into the system prior to study start and throughout the project.

Data from each subject will be recorded in the EDC and will be completed prospectively from patient medical chart by the study staff of each site during or as soon after the visit as possible. All changes made to the clinical data will be captured in an electronic audit trail and available for review by the Sponsor or its representative. CERC will provide clinical monitoring, including review of EDCs with verification to the source documentation per the monitoring plan outlined for the study.

Once all patients complete the study and all queries are answered, the database will be frozen and the analysis will start. Once the study is completed, the records will be kept as requested by local regulations, but in no case less than 15 years from study termination by the sponsor.

The associated e-CRF software and database have been designed to meet regulatory compliance for deployment as part of a validated system compliant with laws and regulations applicable to the conduct of clinical studies pertaining to the use of electronic records and signatures. Database backups are performed regularly. Data Management personnel will employ a full-featured relational database application on a central server that is compliant with to the European General Data Protection Regulation 2016/679 (GDPR).

10. STATEMENT OF COMPLIANCE/ETHICAL AND REGULATORY CONSIDERATIONS

The study will be carried out in accordance to the protocol and principles enunciated in the current version of the Declaration of Helsinki (32), the ISO norm 14155:2011 (33), and applicable national/local regulatory requirements.

The clinical investigational plan, informed consent, any other specific study documents and all amendments to these study documents will be reviewed and approved by the appropriate Ethics Committees (EC) and Competent Authority (CA) before enrolment of any subject. CERC will promptly report to the EC all changes in research activity and all unanticipated problems involving risks to human subjects or others. No changes to the protocol will be made without prior EC approval, except where necessary to eliminate apparent immediate hazards to study participants. Premature study termination or interruption of the study will be reported in accordance to the applicable regulation. The regular end of the study is reported

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to the EC within 90 days, the final study report shall be submitted within one year after study end. Amendments will be reported according to section 12.

11. INFORMED CONSENT

The Investigator or his designees shall ensure that adequate information is given to the subject both in oral and written form, on the nature of the study. Both, the patient information sheet and the consent form will be submitted to the EC to be reviewed and approved in advance.

This information shall be easily understandable by the subject. Information given to the patient shall include the study purpose, the procedures involved, the scale of follow-up examinations, the expected duration as well as the potential risks, benefits and any discomfort participation may entail but also expected benefits for him/her and/or others, risks and inconveniences and an explanation of any alternative methods, and of possible consequences of any withdrawal from the study.

Subjects will also be informed that the participation in the study is voluntary and that he/she may withdraw from the study at any time without reason and that withdrawal of consent will not affect his/her subsequent medical assistance and treatment. Patient shall be allowed sufficient time to decide whether or not they wish to participate. The subjects shall be informed that his/her participation in the clinical investigation is confidential. Subjects shall be made aware that the data relating to the study may be made available to third parties while maintaining anonymity.

Subjects shall sign the informed consent form prior to their inclusion in the study. No clinical investigation specific examination will be performed before written informed consent has been obtained using the approved consent form. A copy of the approved informed consent form along with a copy of each subject's signed consent form will be maintained by each Investigator in a designated clinical study administrative file. A copy of the signed consent form must be given to each subject.

12. AMENDMENTS TO THE CIP

Neither the Sponsor nor the Investigators can modify this clinical investigational plan without obtaining written concurrence from each other and, when appropriate, the site EC. However, all investigators, the EC and the sponsor are allowed to provide suggestions for protocol amendments, however, only the Steering Committee is allowed to change the protocol after approval of the amendments by the EC. The Sponsor will submit the clinical investigational plan modifications, if any, to the appropriate Competent Authorities, as required.

The Sponsor will inform the participating sites about any relevant changes in the CIP. They will be documented as an amendment to the CIP. If applicable, due to the nature of the amendment, and in accordance with local regulations, EC/CA notification/approval is required before the amendment is implemented.

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No changes in the clinical investigation procedures shall be implemented at site before a PI agreement given by signing fully approved amendment is available. Only amendments that are required for patient safety may be implemented prior to regulatory approval.

13. CLINICAL INVESTIGATION PLAN DEVIATIONS

A deviation from the CIP is defined as an event where the clinical investigator or site personnel did not conduct the clinical investigation according to the CIP, standard EN ISO 14155:2011 on clinical investigations with medical devices on human subjects, the latest version of the Declaration of Helsinki as well as any national legal and regulatory applicable requirements.
Any deviations from the clinical investigational plan undertaken to protect the life or physical well-being of a subject in an emergency situation must be reported to CERC within 48 hours of occurrence and if applicable the respective EC as soon as possible, but in no event later than five working days after the emergency occurs.
Major protocol violations include failure to ensure adequate informed consent, recruitment of an ineligible patient into the study on the basis of the inclusion and exclusion criteria and incorrect randomisation of a patient such that they are entered into the wrong treatment arm for non-clinical reasons.
If the major protocol violation is based on inclusion or exclusion criteria, the investigator will be informed of the mistake. During the course of the trial, protocol deviations will be tracked.
Under emergency circumstances, deviations from the protocol to protect the rights, safety and well-being of human subjects may proceed without prior approval of the EC. Such deviations shall be documented and reported to the EC, the sponsor as soon as possible.
The Sponsor or designee is responsible for analysing the deviations and assessing their significance.

14. SUSPENSION OR PREMATURE TERMINATION OF THE CLINICAL INVESTIGATION

The Sponsor, and the Steering Committee will monitor the progress of the study. If warranted, the study may be suspended or discontinued early if there is an observation or a new development presenting an unreasonable risk to the study population.

In the event of study suspension or termination, the Sponsor will send a report outlining the circumstances to the institutional review board (IRB) / EC and all Investigators. A suspended or terminated study may not be reinitiated without approval of the reviewing IRB/EC.

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14.1 STOPPING RULE

In the event of unforeseen or increased risks to subjects encountered during the course of the study, the Sponsor may decide to suspend or terminate the clinical study. Such decision would be made after discussion with the Investigators.

14.2 CONFIDENTIALITY AND DATA PROTECTION

By signing the Patient Informed Consent, the subject authorizes his personal data processing in the framework of the clinical study and is informed of the process for exercising his/her rights (access, rectification, stopping data processing).

Access to patients source documents and e-CRF will be permitted to regulatory authorities and Sponsor's representatives, including Clinical Research Associates, Auditors, Competent Authority or Ethics Committees representatives. All these entities will be granted access to their medical files, for purposes of monitoring, audits and inspections.
The Steering Committee and investigators will have access to the e-CRF and the source data. For the adjudication and classification of primary endpoints, the necessary information will be provided to the CEC in the form of blinded source documents.

These personal data will be managed in a secured and confidential manner taking into account the fact that Sponsor's representatives, including study Committees, Clinical Research Associates and Auditors, ECs/CAs and CERC Data Protection Officer are bound by professional secrecy with regard to all confidential information they have access to, according to GDPR.

15. STUDY RESPONSIBILITIES

15.1 SPONSOR RESPONSIBILITIES

Sponsor has the overall responsibility for its conduct, including assurance that the study meets the regulatory requirements of the Standard ISO 14155 on Clinical Investigations with medical devices on human subjects. Sponsor or its representative will ensure adherence to the ISO 14155 standard and Sponsor’s general duties, selection of Investigators, monitoring, supplemental applications, maintaining records, and submitting periodic and final reports.

The trial Sponsor will adhere to the trial protocol, ISO 14155:2011, GDPR, and compliance with applicable government and institutional regulations. The trial Sponsor is responsible for obtaining proper regulatory approvals and reporting to regulatory authorities per all applicable regulations. Sponsor or its representative will ensure adherence to the ISO 14155:2011 standard and Sponsor’s general duties, selection of Investigators, monitoring, supplemental applications, maintaining records, and submitting periodic and final reports.

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The Sponsor and the CERC will review significant new information, including unanticipated AE, and ensure that such information is provided to the Investigators, and to the Competent Authorities and Ethics Committees, as appropriate.

15.2 INVESTIGATOR RESPONSIBILITIES

The trial Investigator(s) will adhere to the trial protocol, ISO 14155:2011, GDPR, and compliance with applicable government and institutional regulations. The investigator will ensure that the study is conducted in compliance with the CIP and the Investigator's Agreement.

The Investigator will be responsible for the day to day conduct of the clinical investigation as well as for the safety and well-being of the human subjects involved in the clinical investigation.

The Investigator will have the resources to conduct the clinical investigation properly and obtain from the sponsor information which he judges essential about the device and be familiar with this information.

The Investigator must obtain written EC approval prior to including any subject in the study.

The Investigator will be provided by the Sponsor with a sample informed consent that may be modified to meet individual EC (or IRB) requirements. Any modifications of the informed consent document must be submitted to the Sponsor for approval prior to submission to the EC/IRB.

Standardized CRFs to be completed for all enrolled subjects into the study will be provided to the Investigator. Completion of standard CRFs should be accurate and must record of subject's data collected during the study according to ISO 14155 standard and Good Clinical Practices recommendations. It is the responsibility of the Investigator to ensure the quality of the data collected and recorded.

The Investigator will maintain study records for an appropriate time and the subject's identity shall not be released to third parties without the subject's prior consent. Record retention periods will be provided to all concerned by Sponsor. All information and data concerning subjects or their participation in this study will be considered confidential. Only authorized personnel will have access to this confidential information. All data used in the analysis and reporting of this evaluation will be without identifiable reference to individual subjects.

For the purpose of ensuring compliance with the Clinical Investigation Plan, Good Clinical Practices and applicable regulatory requirements, the Investigator should permit auditing by the Sponsor and inspection by applicable Regulatory Authorities. The Investigator agrees to allow the auditors/inspectors to have direct access to his/her study records for review, being understood that these personnel is bound by professional secrecy, and as such will not disclose any personal identity or personal medical information.

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As soon as the Investigator is notified of a future inspection by the Authorities, he/she will inform the Sponsor and authorize the Sponsor to participate in this inspection.

15.3 MONITOR RESPONSIBILITY

The Sponsor has appointed the CERC as Clinical Monitor for this study. CERC personnel are qualified by training and experience to oversee the conduct of the study. CERC will fulfil the responsibilities identified in its standard operating procedures (SOPs), available for review at CERC. These responsibilities include collecting and tracking data forms and trial compliance.

Under the supervision of the Sponsor, monitors will conduct investigational site monitoring to ensure that all Investigators are in compliance with the regulatory requirements of the Standard ISO 14155 and the latest version of the Declaration of Helsinki, with the Clinical Investigation Plan and with the Investigator's Agreement. Sites will be monitored to ensure completed CRFs match the Sponsor records and to resolve any differences. The monitors and Sponsor will evaluate circumstances where an Investigator deviates from the Clinical Investigation Plan.

The Monitor's responsibilities include: maintaining regular contact with each investigational site, through telephone contact and on-site visits, to ensure that the investigational plan is followed; ensure that complete, timely and accurate data are submitted; ensure that problems with inconsistent and incomplete data are addressed; and that the site facilities continue to be adequate. Any questions regarding these matters should be addressed to CERC.

The monitor must ensure that for patients where contact cannot be established, the investigator has attempted to contact the referring physician. In the event that the referring physician has not had contact with the patient, information from the patient's identified point of alternative contact will be utilized. This will also be clearly explained in the Patient Informed Consent form (PIC).
The monitor shall make written report after each visit/contact and be responsible for notifying any deficiencies in writing to the PI and clinical investigation site personnel, providing appropriate and timely corrective actions.

15.4 STUDY COMMITTEES

Description of Study Committees including details on number and kind of members, frequency of meeting, safety information to provide and decision that can be taken with regard to study enrolment/study progress**.**

Steering Commitee

The Steering Committee will be responsible for finalising the protocol, discussing any required amendments, monitoring recruitment rates, ensuring the study runs to time and generally overseeing the running of the study.

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Clinical Event Committee

The Clinical Events Committee - CEC is an independent group of three to four individual members with pertinent expertise that reviews and adjudicates relevant adverse events reported by investigators. The CEC will review a Safety Event dossier, which may include copies of pseudo-anonymised subject source documents provided by clinical investigation sites. CEC members will include experts with the necessary therapeutic and subject matter expertise to adjudicate the event categories outlined above.

The CEC will meet regularly to review and adjudicate all clinical events in which the required minimum data is available. The CEC will also review and rule on all deaths that occur throughout the clinical investigation. All findings of the CEC shall be entered in a specific form on the e-CRF.

CEC responsibilities, qualifications, membership and committee procedures are outlined in the CEC charter.

16. FINANCING

This trial is an investigator-initiated trial supported by CERIC - CEnter for European Research Initiatives in Cardiovascular medicine, Switzerland.

17. PUBLICATION POLICY

All proposed publications and presentations resulting from or relating to the study must be submitted to the steering committee for review and approval prior to submission for publication or presentation.
If the Institution and/or Investigator wants to publish study information in appropriate scientific journals or other professional publications or present such information at scientific conferences/symposia, they may do so under the following conditions: (a) The PI will be able to publish and/or present the data generated from the clinical investigation after agreement with the Sponsor. Data analysis will be done by the statistician and based solely on all available data derived from the e-CRF system after monitoring and validation. (b) only if the publication is consistent with the rules and conventions governing clinical studies and regulatory submissions in all relevant jurisdictions.
The principal authorship of primary publications will be held by the principal investigators.
Co-authorships will be ordered according to the level of contribution, based on the International Committee of Medical Journal Editors (ICJME) guidelines. Exceptions require the prior approval of the Steering Committee. Manuscripts from secondary analyses are anticipated with principal authorship from the Steering Committee or investigators.
Reporting of results will follow guidelines according to the corresponding extension of the CONSORT 2010 statement.(34)

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The Sponsor has the right to include reports of this trial in any regulatory documentation and informational material and will receive a copy of any publication in journals, theses or dissertations.

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18. BIBLIOGRAPHY/REFERENCES

1. Meier B, Gruentzig AR, King SB, Douglas JS, Hollman J, Ischinger T, et al. Risk of side branch occlusion during coronary angioplasty. Am J Cardiol. 1984 Jan 1;53(1):10–4.

2. Iakovou I, Kadota K, Papamentzelopoulos S, Pavlides G, Mitsudo K. Is there a higher risk of stent thrombosis in bifurcation lesion or is it related to the technique? EuroIntervention J Eur Collab Work Group Interv Cardiol Eur Soc Cardiol. 2010 Dec;6 Suppl J:J107-111.

3. Jang WJ, Ahn SG, Song YB, Choi S-H, Chun WJ, Oh JH, et al. Benefit of Prolonged Dual Antiplatelet Therapy After Implantation of Drug-Eluting Stent for Coronary Bifurcation Lesions: Results From the Coronary Bifurcation Stenting Registry II. Circ Cardiovasc Interv. 2018;11(7):e005849.

4. Cho S, Kim J-S, Kang TS, Hong S-J, Shin D-H, Ahn C-M, et al. Long-Term Efficacy of Extended Dual Antiplatelet Therapy After Left Main Coronary Artery Bifurcation Stenting. Am J Cardiol. 2019 Nov 7;

5. Thomas M, Hildick-Smith D, Louvard Y, Albiero R, Darremont O, Stankovic G, et al. Percutaneous coronary intervention for bifurcation disease.A consensus view from the first meeting of the European Bifurcation Club. EuroIntervention J Eur Collab Work Group Interv Cardiol Eur Soc Cardiol. 2006 Aug;2(2):149–53.

6. Zhang F, Dong L, Ge J. Simple versus complex stenting strategy for coronary artery bifurcation lesions in the drug-eluting stent era: a meta-analysis of randomised trials. Heart Br Card Soc. 2009 Oct;95(20):1676–81.

7. Katritsis DG, Siontis GCM, Ioannidis JPA. Double versus single stenting for coronary bifurcation lesions: a meta-analysis. Circ Cardiovasc Interv. 2009 Oct;2(5):409–15.

8. Hildick-Smith D, de Belder AJ, Cooter N, Curzen NP, Clayton TC, Oldroyd KG, et al. Randomized trial of simple versus complex drug-eluting stenting for bifurcation lesions: the British Bifurcation Coronary Study: old, new, and evolving strategies. Circulation. 2010 Mar 16;121(10):1235–43.

9. Gwon H-C, Choi S-H, Song YB, Hahn J-Y, Jeong M-H, Seong I-W, et al. Long-term clinical results and predictors of adverse outcomes after drug-eluting stent implantation for bifurcation lesions in a real-world practice: the COBIS (Coronary Bifurcation Stenting) registry. Circ J Off J Jpn Circ Soc. 2010 Nov;74(11):2322–8.

10. Katritsis DG, Theodorakakos A, Pantos I, Gavaises M, Karcanias N, Efstathopoulos EP. Flow Patterns at Stented Coronary Bifurcations: Computational Fluid Dynamics Analysis. Circ Cardiovasc Interv. 2012 Aug;5(4):530–9.

DocuSign Envelope ID: 2FDB1BA8-CD71-444F-9715-FDD9938BB4E8

11. Behan MW, Holm NR, de Belder AJ, Cockburn J, Erglis A, Curzen NP, et al. Coronary bifurcation lesions treated with simple or complex stenting: 5-year survival from patient-level pooled analysis of the Nordic Bifurcation Study and the British Bifurcation Coronary Study. Eur Heart J. 2016 Jun 21;37(24):1923–8.

12. Lassen JF, Holm NR, Stankovic G, Lefèvre T, Chieffo A, Hildick-Smith D, et al. Percutaneous coronary intervention for coronary bifurcation disease: consensus from the first 10 years of the European Bifurcation Club meetings. EuroIntervention J Eur Collab Work Group Interv Cardiol Eur Soc Cardiol. 2014 Sep;10(5):545–60.

13. Sawaya FJ, Lefèvre T, Chevalier B, Garot P, Hovasse T, Morice M-C, et al. Contemporary Approach to Coronary Bifurcation Lesion Treatment. JACC Cardiovasc Interv. 2016 Sep 26;9(18):1861–78.

14. Sgueglia GA, Chevalier B. Kissing Balloon Inflation in Percutaneous Coronary Interventions. JACC Cardiovasc Interv. 2012 Aug 1;5(8):803–11.

15. Hoye A. The Proximal Optimisation Technique for Intervention of Coronary Bifurcations. Interv Cardiol Lond Engl. 2017 Sep;12(2):110–5.

16. Foin N, Torii R, Mortier P, De Beule M, Viceconte N, Chan PH, et al. Kissing Balloon or Sequential Dilation of the Side Branch and Main Vessel for Provisional Stenting of Bifurcations: Lessons From Micro-Computed Tomography and Computational Simulations. JACC Cardiovasc Interv. 2012 Jan 1;5(1):47–56.

17. Finet G, Derimay F, Motreff P, Guerin P, Pilet P, Ohayon J, et al. Comparative Analysis of Sequential Proximal Optimizing Technique Versus Kissing Balloon Inflation Technique in Provisional Bifurcation Stenting: Fractal Coronary Bifurcation Bench Test. JACC Cardiovasc Interv. 2015 Aug 24;8(10):1308–17.

18. Zhong M, Tang B, Zhao Q, Cheng J, Jin Q, Fu S. Should kissing balloon inflation after main vessel stenting be routine in the one-stent approach? A systematic review and meta-analysis of randomized trials. PloS One. 2018;13(6):e0197580.

19. Liu G, Ke X, Huang Z-B, Wang L-C, Huang Z-N, Guo Y, et al. Final kissing balloon inflation for coronary bifurcation lesions treated with single-stent technique : A meta-analysis. Herz. 2019 Jun;44(4):354–62.

20. Niemelä M, Kervinen K, Erglis A, Holm NR, Maeng M, Christiansen EH, et al. Randomized comparison of final kissing balloon dilatation versus no final kissing balloon dilatation in patients with coronary bifurcation lesions treated with main vessel stenting: the Nordic-Baltic Bifurcation Study III. Circulation. 2011 Jan 4;123(1):79–86.

21. Pan M, Medina A, Suárez de Lezo J, Romero M, Segura J, Martín P, et al. Coronary bifurcation lesions treated with simple approach (from the Cordoba & Las Palmas [CORPAL] Kiss Trial). Am J Cardiol. 2011 May 15;107(10):1460–5.

DocuSign Envelope ID: 2FDB1BA8-CD71-444F-9715-FDD9938BB4E8

22. Gwon H-C, Hahn J-Y, Koo B-K, Song YB, Choi S-H, Choi J-H, et al. Final kissing ballooning and long-term clinical outcomes in coronary bifurcation lesions treated with 1-stent technique: results from the COBIS registry. Heart Br Card Soc. 2012 Feb;98(3):225–31.

23. Yamawaki M, Muramatsu T, Kozuma K, Ito Y, Kawaguchi R, Kotani J-I, et al. Long-term clinical outcome of a single stent approach with and without a final kissing balloon technique for coronary bifurcation. Circ J Off J Jpn Circ Soc. 2014;78(1):110–21.

24. Biondi-Zoccai G, Sheiban I, De Servi S, Tamburino C, Sangiorgi G, Romagnoli E. To kiss or not to kiss? Impact of final kissing-balloon inflation on early and long-term results of percutaneous coronary intervention for bifurcation lesions. Heart Vessels. 2014 Nov;29(6):732–42.

25. U.S. Food and Drug Administration. Medtronic Resolute Onyx Stent - Premarket Approval (PMA) [Internet]. 2017. Available from: https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpma/pma.cfm?id=P160043

26. Medtronic. Medtronic - Resolute Onyx - Instructions for Use - M981345A001 Rev. 1A [Internet]. 2018. Available from: http://manuals.medtronic.com/content/dam/emanuals/cardio/M981345A001_Rev1A_DOC1_view.pdf

27. Garcia-Garcia HM, McFadden EP, Farb A, Mehran R, Stone GW, Spertus J, et al. Standardized End Point Definitions for Coronary Intervention Trials: The Academic Research Consortium-2 Consensus Document. Circulation. 2018 12;137(24):2635–50.

28. Campeau L. Letter: Grading of angina pectoris. Circulation. 1976 Sep 1;54(3):522–3.

29. Murasato Y, Kinoshita Y, Yamawaki M, Shinke T, Otake H, Takeda Y, et al. Efficacy of kissing balloon inflation after provisional stenting in bifurcation lesions guided by intravascular ultrasound: short and midterm results of the J-REVERSE registry. EuroIntervention J Eur Collab Work Group Interv Cardiol Eur Soc Cardiol. 2016 Feb;11(11):e1237-1248.

30. Levine GN, Bates ER, Blankenship JC, Bailey SR, Bittl JA, Cercek B, et al. 2011 ACCF/AHA/SCAI Guideline for Percutaneous Coronary Intervention: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines and the Society for Cardiovascular Angiography and Interventions. Circulation. 2011 Dec 6;124(23):e574-651.

31. Neumann F-J, Sousa-Uva M, Ahlsson A, Alfonso F, Banning AP, Benedetto U, et al. 2018 ESC/EACTS Guidelines on myocardial revascularization. Eur Heart J. 2019 Jan 7;40(2):87–165.

32. The World Medical Association. Declaration of Helsinki – Ethical Principles for Medical Research Involving Human Subjects [Internet]. 2018 [cited 2020 Jan 3]. Available from:

DocuSign Envelope ID: 2FDB1BA8-CD71-444F-9715-FDD9938BB4E8

https://www.wma.net/policies-post/wma-declaration-of-helsinki-ethical-principles-for-medical-research-involving-human-subjects/

33. ISO. Clinical investigation of medical devices for human subjects - Good clinical practice - ISO 14155:2011 [Internet]. ISO. 2011 [cited 2020 Jan 3]. Available from: http://www.iso.org/cms/render/live/en/sites/isoorg/contents/data/standard/04/55/45557.html

34. Piaggio G, Elbourne DR, Pocock SJ, Evans SJW, Altman DG, CONSORT Group. Reporting of noninferiority and equivalence randomized trials: extension of the CONSORT 2010 statement. JAMA. 2012 Dec 26;308(24):2594–604.

35. The World Medical Association. Declaration of Helsinki – Ethical Principles for Medical Research Involving Human Subjects [Internet]. 2018 [cited 2020 Jan 22]. Available from: https://www.wma.net/policies-post/wma-declaration-of-helsinki-ethical-principles-for-medical-research-involving-human-subjects/

36. Mehran R, Rao SV, Bhatt DL, Gibson CM, Caixeta A, Eikelboom J, et al. Standardized bleeding definitions for cardiovascular clinical trials: a consensus report from the Bleeding Academic Research Consortium. Circulation. 2011 Jun 14;123(23):2736–47.

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19. APPENDICES

19.1 APPENDIX I: PARTICIPATING INVESTIGATORS AND INVESTIGATION SITES

LIST OF PARTICIPATING INVESTIGATORS AND INVESTIGATION SITES

Country	Name	PI name	Address
France	**Toulouse**	Dr. Frédéric Bouisset	CHU Toulouse Rangueil Service de Cardiologie B 1, avenue du Pr Jean Poulhès 31959 Toulouse Cedex 9
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France	**Quincy-Sous-Sénart**	Dr. Stephane Champagne	Hôpital Privé Claude Galien ICPS 20, route de Boussy-Saint-Antoine 91480 Quincy-Sous-Sénart
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France	**Marseille**	Prof. Thomas Cuisset	CHU Timone Adultes Service de cardiologie du Pr Bonnet, 10ème étage 264 Rue Saint Pierre 13385 Marseille
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Switzerland	**Lausanne**	Dr. Vladimir Rubimbura	CHUV Centre Hospitalier Universitaire Vaudois Service de Cardiologie Bâtiment P2-bureau 113 Av. de Beaumont 21bis CH-1011 Lausanne
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Spain	**Madrid**	Dr. Maria Teresa Velázquez Martín	Hospital Universitario 12 de Octubre Avenida de Córdoba, s/n 28041 Madrid
Spain	**Cordoba**	Dr. Manuel Pan	Hospital Universitario Reina Sofia vda. Menéndez Pidal, s/n 14004 Córdoba
Spain	**Girona**	Dr. Juan Rondan	Hospital de Cabuenes c/ Los Prados 395 33203 Gijon

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19.2 APPENDIX II: DECLARATION OF HELSINKI

WMA Declaration of Helsinki (35)

Ethical Principles for Medical Research Involving Human Subjects

Adopted by the 18th WMA General Assembly, Helsinki, Finland, June 1964
and amended by the:
29th WMA General Assembly, Tokyo, Japan, October 1975
35th WMA General Assembly, Venice, Italy, October 1983
41st WMA General Assembly, Hong Kong, September 1989
48th WMA General Assembly, Somerset West, Republic of South Africa, October 1996
52nd WMA General Assembly, Edinburgh, Scotland, October 2000
53rd WMA General Assembly, Washington DC, USA, October 2002 (Note of Clarification added)
55th WMA General Assembly, Tokyo, Japan, October 2004 (Note of Clarification added)
59th WMA General Assembly, Seoul, Republic of Korea, October 2008
64th WMA General Assembly, Fortaleza, Brazil, October 2013

Preamble

1. The World Medical Association (WMA) has developed the Declaration of Helsinki as a statement of ethical principles for medical research involving human subjects, including research on identifiable human material and data.
   The Declaration is intended to be read as a whole and each of its constituent paragraphs should be applied with consideration of all other relevant paragraphs.
2. Consistent with the mandate of the WMA, the Declaration is addressed primarily to physicians. The WMA encourages others who are involved in medical research involving human subjects to adopt these principles.

General Principles

3. The Declaration of Geneva of the WMA binds the physician with the words, “The health of my patient will be my first consideration,” and the International Code of Medical Ethics declares that, “A physician shall act in the patient's best interest when providing medical care.”
4. It is the duty of the physician to promote and safeguard the health, well-being and rights of patients, including those who are involved in medical research. The physician's knowledge and conscience are dedicated to the fulfillment of this duty.
5. Medical progress is based on research that ultimately must include studies involving human subjects.
6. The primary purpose of medical research involving human subjects is to understand the causes, development and effects of diseases and improve preventive, diagnostic and therapeutic interventions (methods, procedures and treatments). Even the best proven interventions must be

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evaluated continually through research for their safety, effectiveness, efficiency, accessibility and quality.

7. Medical research is subject to ethical standards that promote and ensure respect for all human subjects and protect their health and rights.
8. While the primary purpose of medical research is to generate new knowledge, this goal can never take precedence over the rights and interests of individual research subjects.
9. It is the duty of physicians who are involved in medical research to protect the life, health, dignity, integrity, right to self-determination, privacy, and confidentiality of personal information of research subjects. The responsibility for the protection of research subjects must always rest with the physician or other health care professionals and never with the research subjects, even though they have given consent.
10. Physicians must consider the ethical, legal and regulatory norms and standards for research involving human subjects in their own countries as well as applicable international norms and standards. No national or international ethical, legal or regulatory requirement should reduce or eliminate any of the protections for research subjects set forth in this Declaration.
11. Medical research should be conducted in a manner that minimizes possible harm to the environment.
12. Medical research involving human subjects must be conducted only by individuals with the appropriate ethics and scientific education, training and qualifications. Research on patients or healthy volunteers requires the supervision of a competent and appropriately qualified physician or other health care professional.
13. Groups that are underrepresented in medical research should be provided appropriate access to participation in research.
14. Physicians who combine medical research with medical care should involve their patients in research only to the extent that this is justified by its potential preventive, diagnostic or therapeutic value and if the physician has good reason to believe that participation in the research study will not adversely affect the health of the patients who serve as research subjects.
15. Appropriate compensation and treatment for subjects who are harmed as a result of participating in research must be ensured.

Risks, Burdens and Benefits

16. In medical practice and in medical research, most interventions involve risks and burdens.
    Medical research involving human subjects may only be conducted if the importance of the objective outweighs the risks and burdens to the research subjects.
17. All medical research involving human subjects must be preceded by careful assessment of predictable risks and burdens to the individuals and groups involved in the research in comparison with foreseeable benefits to them and to other individuals or groups affected by the condition under investigation.
    Measures to minimize the risks must be implemented. The risks must be continuously monitored, assessed and documented by the researcher.
18. Physicians may not be involved in a research study involving human subjects unless they are confident that the risks have been adequately assessed and can be satisfactorily managed.
    When the risks are found to outweigh the potential benefits or when there is conclusive proof of definitive outcomes, physicians must assess whether to continue, modify or immediately stop the study.

Vulnerable Groups and Individuals

19. Some groups and individuals are particularly vulnerable and may have an increased likelihood of being wronged or of incurring additional harm.
    All vulnerable groups and individuals should receive specifically considered protection.

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20. Medical research with a vulnerable group is only justified if the research is responsive to the health needs or priorities of this group and the research cannot be carried out in a non-vulnerable group. In addition, this group should stand to benefit from the knowledge, practices or interventions that result from the research.

Scientific Requirements and Research Protocols

21. Medical research involving human subjects must conform to generally accepted scientific principles, be based on a thorough knowledge of the scientific literature, other relevant sources of information, and adequate laboratory and, as appropriate, animal experimentation. The welfare of animals used for research must be respected.
22. The design and performance of each research study involving human subjects must be clearly described and justified in a research protocol.
    The protocol should contain a statement of the ethical considerations involved and should indicate how the principles in this Declaration have been addressed. The protocol should include information regarding funding, sponsors, institutional affiliations, potential conflicts of interest, incentives for subjects and information regarding provisions for treating and/or compensating subjects who are harmed as a consequence of participation in the research study.
    In clinical trials, the protocol must also describe appropriate arrangements for post-trial provisions.

Research Ethics Committees

23. The research protocol must be submitted for consideration, comment, guidance and approval to the concerned research ethics committee before the study begins. This committee must be transparent in its functioning, must be independent of the researcher, the sponsor and any other undue influence and must be duly qualified. It must take into consideration the laws and regulations of the country or countries in which the research is to be performed as well as applicable international norms and standards but these must not be allowed to reduce or eliminate any of the protections for research subjects set forth in this Declaration.
    The committee must have the right to monitor ongoing studies. The researcher must provide monitoring information to the committee, especially information about any serious adverse events. No amendment to the protocol may be made without consideration and approval by the committee. After the end of the study, the researchers must submit a final report to the committee containing a summary of the study's findings and conclusions.

Privacy and Confidentiality

24. Every precaution must be taken to protect the privacy of research subjects and the confidentiality of their personal information.

Informed Consent

25. Participation by individuals capable of giving informed consent as subjects in medical research must be voluntary. Although it may be appropriate to consult family members or community leaders, no individual capable of giving informed consent may be enrolled in a research study unless he or she freely agrees.
26. In medical research involving human subjects capable of giving informed consent, each potential subject must be adequately informed of the aims, methods, sources of funding, any possible conflicts of interest, institutional affiliations of the researcher, the anticipated benefits and potential risks of the

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study and the discomfort it may entail, post-study provisions and any other relevant aspects of the study. The potential subject must be informed of the right to refuse to participate in the study or to withdraw consent to participate at any time without reprisal. Special attention should be given to the specific information needs of individual potential subjects as well as to the methods used to deliver the information.

After ensuring that the potential subject has understood the information, the physician or another appropriately qualified individual must then seek the potential subject's freely-given informed consent, preferably in writing. If the consent cannot be expressed in writing, the non-written consent must be formally documented and witnessed.

All medical research subjects should be given the option of being informed about the general outcome and results of the study.

27. When seeking informed consent for participation in a research study the physician must be particularly cautious if the potential subject is in a dependent relationship with the physician or may consent under duress. In such situations the informed consent must be sought by an appropriately qualified individual who is completely independent of this relationship.
28. For a potential research subject who is incapable of giving informed consent, the physician must seek informed consent from the legally authorized representative. These individuals must not be included in a research study that has no likelihood of benefit for them unless it is intended to promote the health of the group represented by the potential subject, the research cannot instead be performed with persons capable of providing informed consent, and the research entails only minimal risk and minimal burden.
29. When a potential research subject who is deemed incapable of giving informed consent is able to give assent to decisions about participation in research, the physician must seek that assent in addition to the consent of the legally authorized representative. The potential subject's dissent should be respected.
30. Research involving subjects who are physically or mentally incapable of giving consent, for example, unconscious patients, may be done only if the physical or mental condition that prevents giving informed consent is a necessary characteristic of the research group. In such circumstances the physician must seek informed consent from the legally authorized representative. If no such representative is available and if the research cannot be delayed, the study may proceed without informed consent provided that the specific reasons for involving subjects with a condition that renders them unable to give informed consent have been stated in the research protocol and the study has been approved by a research ethics committee. Consent to remain in the research must be obtained as soon as possible from the subject or a legally authorized representative.
31. The physician must fully inform the patient which aspects of their care are related to the research. The refusal of a patient to participate in a study or the patient's decision to withdraw from the study must never adversely affect the patient-physician relationship.
32. For medical research using identifiable human material or data, such as research on material or data contained in biobanks or similar repositories, physicians must seek informed consent for its collection, storage and/or reuse. There may be exceptional situations where consent would be impossible or impracticable to obtain for such research. In such situations the research may be done only after consideration and approval of a research ethics committee.

Use of Placebo

33. The benefits, risks, burdens and effectiveness of a new intervention must be tested against those of the best proven intervention(s), except in the following circumstances:
    Where no proven intervention exists, the use of placebo, or no intervention, is acceptable; or
    Where for compelling and scientifically sound methodological reasons the use of any intervention less effective than the best proven one, the use of placebo, or no intervention is necessary to determine the efficacy or safety of an intervention

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and the patients who receive any intervention less effective than the best proven one, placebo, or no intervention will not be subject to additional risks of serious or irreversible harm as a result of not receiving the best proven intervention.

Extreme care must be taken to avoid abuse of this option.

Post-Trial Provisions

34. In advance of a clinical trial, sponsors, researchers and host country governments should make provisions for post-trial access for all participants who still need an intervention identified as beneficial in the trial. This information must also be disclosed to participants during the informed consent process.

Research Registration and Publication and Dissemination of Results

35. Every research study involving human subjects must be registered in a publicly accessible database before recruitment of the first subject.
36. Researchers, authors, sponsors, editors and publishers all have ethical obligations with regard to the publication and dissemination of the results of research. Researchers have a duty to make publicly available the results of their research on human subjects and are accountable for the completeness and accuracy of their reports. All parties should adhere to accepted guidelines for ethical reporting. Negative and inconclusive as well as positive results must be published or otherwise made publicly available. Sources of funding, institutional affiliations and conflicts of interest must be declared in the publication. Reports of research not in accordance with the principles of this Declaration should not be accepted for publication.

Unproven Interventions in Clinical Practice

37. In the treatment of an individual patient, where proven interventions do not exist or other known interventions have been ineffective, the physician, after seeking expert advice, with informed consent from the patient or a legally authorized representative, may use an unproven intervention if in the physician's judgment it offers hope of saving life, re-establishing health or alleviating suffering. This intervention should subsequently be made the object of research, designed to evaluate its safety and efficacy. In all cases, new information must be recorded and, where appropriate, made publicly available.

19.3 APPENDIX III: TERMS AND DEFINITIONS

Term	Definition
ACUTE VESSEL OCCLUSION	The state of complete luminal obstruction with no antegrade blood flow

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Term	Definition
ANGINA STATUS (CCS (28))	**Grade 1:** "Ordinary physical activity does not cause ... angina," such as walking and climbing stairs. Angina with strenuous or rapid or prolonged exertion at work or recreation. **Grade 2:** "Slight limitation of ordinary activity." Walking or climbing stairs rapidly, walking uphill, walking or stair climbing after meals, or in cold, or in wind, or under emotional stress, or only during the few hours after awakening. Walking more than 2 blocks on the level and climbing more than one flight of ordinary stairs at a normal pace and in normal conditions. **Grade 3:** "Marked limitation of ordinary physical activity." Walking one to two blocks on the level and clim-bing one flight of stairs in normal conditions and at normal pace. **Grade 4:** "Inability to carry on any physical activity without discomfort - anginal syndrome may be present at rest."
BLEEDING (BARC (36))	**Life-threatening or Disabling Bleeding** • Fatal bleeding (Bleeding Academic Research Consortium [BARC] type 5) OR • Bleeding in a critical organ, such as intracranial, intraspinal, intraocular, or pericardial necessitating pericardiocentesis, or intramuscular with compartment syndrome (BARC type 3b and 3c) OR • Bleeding causing hypovolemic shock or severe hypotension requiring vasopressors or surgical intervention (BARC type 3b) OR • Overt source of bleeding with drop in hemoglobin of ≥5 g/dL or whole blood or packed red blood cells (RBC) transfusion ≥4 units (BARC type 3b)* **Major Bleeding (BARC type 3a)** • Overt bleeding either associated with a drop in the hemoglobin level of at least 3.0g/dL or requiring transfusion of 2 or 3 units of whole blood/RBC, or causing hospitalization or permanent injury, or requiring surgical intervention AND does not meet criteria of life-threatening or disabling bleeding **Minor Bleeding (BARC type 2 or 3a, depending on the severity)** • Any bleeding worthy of clinical mention (e.g., access site hematoma) that does not qualify as life-threatening, disabling, or major ** Given one unit of packed RBC typically will raise blood hemoglobin concentration by 1 g/dL, an estimated decrease in hemoglobin will be calculated.*

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CARDIAC DECOMPENSATION	Inability of the heart to maintain adequate circulation.
CARDIOGENIC SHOCK	An insufficient forward cardiac output to maintain adequate perfusion of vital organs to meet ongoing demands for oxygenation and metabolism. Cardiogenic shock is due to either inadequate left ventricular pump function (such as in congestive heart failure) or inadequate left ventricular filling (such as in cardiac tamponade). Cardiogenic shock is defined as sustained hypotension (>30 minutes) with evidence of tissue hypoperfusion including oliguria (<30 mL/h), cool extremities, cyanosis, and altered mental status.
CEREBRAL INFARCTION	Evidence of brain cell death from imaging studies or pathological examination. If there are clinical symptoms, then it is a stroke; otherwise, it is an asymptomatic cerebral infarction.
CORONARY OBSTRUCTION	Angiographic or echocardiographic evidence of a new, partial or complete, obstruction of a coronary ostium, by the native leaflets, calcifications, or dissection, occurring during or after the procedure. The diagnosis of coronary obstruction can be determined by imaging studies (coronary angiography, intravascular ultrasound, multi-slice computed tomography angiography, or echocardiography), surgical exploration, or autopsy findings. Cardiac biomarker elevations and ECG changes indicating new ischemia provide corroborative evidence.
DEATH (ARC-2 (27))	**Cardiovascular death** Defined as death resulting from cardiovascular causes. The following categories may be collected: 1. Death caused by acute MI 2. Death caused by sudden cardiac, including unwitnessed, death 3. Death resulting from heart failure 4. Death caused by stroke 5. Death caused by cardiovascular procedures 6. Death resulting from cardiovascular hemorrhage 7. Death resulting from other cardiovascular Cause **Noncardiovascular death:** Defined as any death that is not thought to be the result of a

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	cardiovascular cause. The following categories may be collected: 1. Death resulting from malignancy 2. Death resulting from pulmonary causes 3. Death caused by infection (includes sepsis) 4. Death resulting from gastrointestinal causes 5. Death resulting from accident/trauma 6. Death caused by other noncardiovascular organ failure 7. Death resulting from other noncardiovascular cause **Undetermined** Defined as a death not attributable to any other category because of the absence of any relevant source documents. Such deaths will be classified as cardiovascular for end point determination.
MYOCARDIAL INFARCTION (MI) (ARC-2) (27)	**Criteria for acute myocardial infarction** The term acute myocardial infarction (MI) should be used when there is evidence of myocardial necrosis in a clinical setting consistent with acute myocardial ischemia. Under these conditions any one of the following criteria meets the diagnosis for MI: ● Detection of a rise and/or fall of cardiac biomarker values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile upper reference limit (URL) and with at least one of the following: - Symptoms of ischemia. - New or presumed new significant ST-segment–T wave (ST–T) changes or new left bundle branch block (LBBB). - Development of pathological Q waves in the ECG. - Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality. - Identification of an intracoronary thrombus by angiography or autopsy. ● Cardiac death with symptoms suggestive of myocardial ischemia and presumed new ischemic ECG changes or new LBBB, but death occurred before cardiac biomarkers were obtained, or before cardiac biomarker values would be increased. ● Percutaneous coronary intervention (PCI) related MI is arbitrarily defined by elevation of cTn values (>5 x 99th percentile URL) in patients with normal baseline values (≤99th percentile URL) or a rise of cTn values >20% if the baseline values are elevated and are stable or falling. In addition, either (i) symptoms

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	suggestive of myocardial ischemia or (ii) new ischemic ECG changes or (iii) angiographic findings consistent with a procedural complication or (iv) imaging demonstration of new loss of viable myocardium or new regional wall motion abnormality are required. ● Stent thrombosis associated with MI when detected by coronary angiography or autopsy in the setting of myocardial ischemia and with a rise and/or fall of cardiac biomarker values with at least one value above the 99th percentile URL. ● Coronary artery bypass grafting (CABG) related MI is arbitrarily defined by elevation of cardiac biomarker values (>10 x 99th percentile URL) in patients with normal baseline cTn values (≤99th percentile URL). In addition, either (i) new pathological Q waves or new LBBB, or (ii) angiographic documented new graft or new native coronary artery occlusion, or (iii) imaging evidence of new loss of viable myocardium or new regional wall motion abnormality. **Criteria for prior myocardial infarction** Any one of the following criteria meets the diagnosis for prior MI: ● Pathological Q waves with or without symptoms in the absence of non-ischemic causes. ● Imaging evidence of a region of loss of viable myocardium that is thinned and fails to contract, in the absence of a non-ischemic cause. ● Pathological findings of a prior MI.
PERI-PROCEDURAL MI (ARC-2) (27)	Absolute rise in cardiac troponin (from baseline) ≥35 times upper reference limit + 1 (or more) of the following criteria: - New significant* Q waves or equivalent - Flow-limiting angiographic complications - New “substantial” loss of myocardium on imaging **Significant periprocedural myocardial injury:** Absolute rise in cardiac troponin (from baseline) ≥70 times upper reference limit *Definitions applicable in patients with normal (or elevated and stable or falling) baseline biomarkers. *Q-wave criteria requires the development of new Q waves ≥40 ms in duration and ≥1 mm deep in voltage in ≥2 contiguous leads*

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Term	Definition
PROCEDURE-RELATED EVENTS	Events occurring during or as a direct result of the index procedure.
SOURCE DATA (per ISO 14155:2011)	All information in original records of clinical findings, observations, or other activities in a clinical investigation, necessary for the reconstruction and evaluation of the clinical investigation.
SOURCE DOCUMENT (per ISO 14155:2011)	Printed, optical or electronic document containing source data. Examples: Hospital records, laboratory notes, device accountability records, photographic negatives, radiographs, records kept at the investigation center, at the laboratories and at the medico-technical departments involved in the clinical investigation.
STENT THROMBOSIS (ARC-2) (27)	**Definite stent thrombosis:** Angiographic confirmation of stent thrombosis* • The presence of a thrombus† that originates in the stent/scaffold or in the segment 5 mm proximal or distal to the stent/scaffold or in a side branch originating from the stented segment and the presence of at least 1 of the following criteria: - Acute onset of ischemic symptoms at rest, - New electrocardiographic changes suggestive of acute ischemia, - Typical rise and fall in cardiac biomarkers (refer to definition of spontaneous myocardial infarction), OR • Pathological confirmation of stent thrombosis - Evidence of recent thrombus within the stent determined at autopsy - Examination of tissue retrieved following thrombectomy (visual/histology) **Probable stent thrombosis:** Regardless of the time after the index procedure, any myocardial infarction that is related to documented acute ischemia in the territory of the implanted stent/scaffold without angiographic confirmation of stent thrombosis and in the absence of any other obvious cause.‡ **Silent stent occlusion:** The incidental angiographic documentation of stent occlusion in the absence of clinical signs or symptoms is not considered stent thrombosis. **Timing of Stent thrombosis** (duration after stent implantation):

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	- Acute: 0§–24 hours - Subacute: >24 h–30 days - Late: >30 d–1 year - Very late: >1 year *Early stent thrombosis is 0 to 30 days (acute plus subacute stent thrombosis).* **Definite stent/scaffold thrombosis is considered to have occurred by either angiographic or pathological confirmation.* *†Occlusive thrombus: Thrombolysis in Myocardial Infarction grade 0 or 1 flow within or proximal to a stent/scaffold segment. Nonocclusive thrombus: intracoronary thrombus is defined as a (spherical, ovoid, or irregular) noncalcified filling defect or lucency surrounded by contrast material (on 3 sides or within a coronary stenosis) seen in multiple projections, persistence of contrast material within the lumen, or visible embolization of intraluminal material downstream.* *‡When the stented/scaffolded segment is in the left circumflex coronary artery or in the presence of preexisting electrocardiographic abnormalities (eg, left bundle branch block, paced rhythms), definitive evidence of localization may be absent and Clinical Events Committee adjudication is based on review of all available evidence).* *§Defined as the moment the patient is undraped and taken off the catheterization table.*
STROKE	Sudden onset of vertigo, numbness, dysphasia, weakness, visual field defects, dysarthria or other focal neurological deficits due to vascular lesions of the brain such as hemorrhage, embolism, thrombosis, or rupturing aneurysm, that persists >24 hours.
TARGET LESION	Treated lesion starting 5 mm proximal of the stented lesion and to end 5 mm distal of the stented lesion.
TARGET LESION FAILURE	Composite of: Cardiac Death, Target vessel Myocardial Infarction, Clinically driven Target lesion revascularization
TARGET LESION RE-VASCULARIZATION	Any repeat percutaneous intervention of the target lesion or bypass surgery of the target vessel. Clinically-driven revascularizations are those in which the patient has a positive functional study, ischemic ECG changes at rest in a distribution consistent with the target vessel, or ischemic symptoms. Revascularization of a target lesion with an in-lesion diameter stenosis ≥70% (by QCA) in the absence of the above-mentioned ischemic signs or symptoms is also considered clinically-driven. In the absence of QCA data for relevant follow-up angiograms, the clinical need for revascularization is adjudicated using the presence or absence of ischemic signs and symptoms. Non-clinically driven repeat target lesion revascularizations are those in which the patient undergoes a non-emergent

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	revascularization for a diameter stenosis <50% (by QCA). Non-emergent repeat target lesion revascularization for a diameter stenosis <70% (by QCA) in patients without either a positive functional study or angina are also considered non-clinically driven defined as any repeat revascularization of the target site whether by PCI or bypass surgery.
TARGET VESSEL	Index coronary artery which was in physical contact with any component (guiding catheter, guide wire, balloon catheter, etc.) of the angioplasty hardware during the initial procedure.
TARGET VESSEL RE-VASCULARIZATION	Any target vessel revascularization, death, or MI attributed to the target vessel.